Survey-Native Clustering Statistics on the Lightcone: A Unified Four-Flavor Framework in (θ, z_1, z_2)

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Abstract

We argue that the natural data primitive for clustering analyses on redshift surveys is the set of four-flavor pair counts and counts-in-cells moments $\{N^p\}^{\text{flavor}}(\theta; z_1, z_2)$ indexed by angular separation θ, two endpoint redshifts (z_1, z_2), moment order p, and flavor $\in \{DD, DR, RD, RR\}$. The flavors index which side of the pair (or cell) is data vs. reference, where references are angular random samples at user-chosen redshifts. From this primitive, traditional clustering observables emerge as Landy–Szalay, Hamilton, or natural- estimator combinations of flavor moments. The same primitive admits an equivalent kNN-CDF representation via the Banerjee–Abel relation, providing a complete one-point counts-in-cells PDF at each (θ, z_1, z_2) triple. We show that traditional 2-point multipoles $\xi_\ell(s)$ can be derived from the diagonal $(z_1 = z_2)$ subset by a known cosmology- dependent projection, demonstrating that the framework loses no information relative to standard analyses while keeping cosmology out of the data vector. Both DD and RD flavors are required for full information extraction; halo mass thresholds, photometric redshift uncertainties, and the lightcone evolution of all observables are natural extensions of the core formalism.

1 Motivation

In a redshift survey the directly observed quantities for each galaxy are two angles (α, δ) and a redshift z. Conventional analyses convert these to a 3D position $\mathbf{x}$ under an assumed background cosmology, then compute pair statistics in the resulting Euclidean space. This procedure has three structural problems:

1. *Cosmology contaminates the data vector.* The Alcock–Paczynski distortion is the most visible consequence: an incorrect fiducial $D_M(z)$ or $H(z)$ produces a geometric distortion of the inferred 3D field that is degenerate with redshift-space distortions and the clustering signal.

2. *Mixed observables.* A 3D pair separation r combines an angular separation (an observable) with a radial separation (a derived quantity). The two enter measurement errors and selection effects very differently.

3. *Implicit redshift averaging.* A 3D two-point function $\xi(r)$ on a survey with broad redshift coverage is a kernel-weighted average of $\xi(r; z)$ over the lightcone, with weighting set by the survey $\bar{n}(z)$. This averaging is rarely written down explicitly, hidden in the choice of effective redshift.

We propose a reformulation in which the data primitive is the set of four-flavor counts on $(\log\theta, \log(1+z_1), \log(1+z_2))$, constructed from angular pair counting and reference redshift anchors. Cosmology enters only at the forward-modeling stage. From a single pipeline pass, all standard clustering observables follow as Landy–Szalay or Hamilton combinations of flavor moments. We argue this is the natural unification of pair-counting and counts-in-cells approaches and that it loses no information relative to conventional analyses.

2 The data primitive

2.1 Reference redshifts and the four flavors

The framework requires two catalogs and a set of reference points:

- The *data catalog* $\{(\Omega_i^D, z_i^D)\}$: actual galaxies with observed sky positions and redshifts.
- The *reference catalog* $\{(\Omega_j^R, z_q^{(r)})\}$: angular sampling points (e.g., HEALPix pixel centers, or Poisson-sampled uniform on the survey footprint) at a discrete set of user-chosen *reference redshifts* $\{z_q^{(r)}\}_{r=1}^{N_{\rm ref}}$. Each reference point sits at exactly one of the chosen redshifts; no redshift uncertainty is assigned. The reference catalog samples the angular footprint and selection mask of the survey, replicated at each reference redshift.

The reference redshifts are theoretical anchors chosen by the analyst, typically on a uniform $\log(1+z)$ grid, or denser around specific features of interest (BAO scale, RSD-sensitive epochs). The reference catalog has no intrinsic redshift distribution — it is built to be exact at each chosen $z_q^{(r)}$.

A pair (P_1, P_2) of points (one from each catalog or two from the same) contributes to the data primitive based on which catalog each member comes from:

P_1	P_2	Flavor	Records (θ_{12}, z_1, z_2) as
D	D	DD	$(\theta_{12}, z_1^D, z_2^D)$
D	R	DR	$(\theta_{12}, z_1^D, z_q^{(r)})$ for the R-side reference
R	D	RD	$(\theta_{12}, z_q^{(r)}, z_2^D)$ for the R-side reference
R	R	RR	$(\theta_{12}, z_q^{(r)}, z_q^{(r')})$ for both R-side references

A key property: in the DD case, both redshift labels are intrinsic to the galaxies and the reference grid is not involved. The reference redshifts appear only on the R-side of any pair. Consequently, DD pair counts are reference-grid-independent — they characterize the galaxy distribution directly. The DR, RD, and RR counts inherit their redshift structure from the chosen reference grid.

2.2 Pair-count primitive

For each flavor and each (θ, z_1, z_2) bin, the primary observable is the pair count

$$n_{\rm pairs}^{\rm flavor}(\theta; z_1, z_2) = \#\big\{(P_1, P_2) : \theta_{12} \in [\theta - d\theta/2, \theta + d\theta/2],\ z_1, z_2 \text{ in bin}\big\}. \tag{1}$$

For continuous endpoints (galaxy redshifts in DD, one side of DR and RD), z_1 or z_2 is binned at the resolution of the reference grid or finer. For reference endpoints, the bin reduces to a delta function at the reference grid value.

In each cell of the (θ, z_1, z_2) grid, the four pair counts $\{DD, DR, RD, RR\}$ together carry the information about the angular clustering between populations selected at the two endpoint redshifts.

2.3 Moment primitive: counts-in-cells per flavor

The counts-in-cells representation is the one-point analog. For each flavor, define the count of P_2-type points within an angular cap of radius θ around a P_1-type center, with the P_2-points restricted to the redshift bin z_2:

$$N^{\text{flavor}}(\theta; z_2 \mid P_1 \text{ at } z_1) = \#\big\{P_2 : \theta_{P_1 P_2} \le \theta,\ z_{P_2} \text{ in bin } z_2\big\}. \tag{2}$$

Averaging over P_1-centers in bin z_1 produces the per-flavor moment

$$\langle N^p \rangle^{\text{flavor}}(\theta; z_1, z_2) = \big\langle N^{\text{flavor}}(\theta; z_2 \mid P_1 \text{ at } z_1)^p \big\rangle_{P_1 \in z_1}. \tag{3}$$

For $p = 1$ this is the mean count; for $p = 2$ it gives the variance via $\sigma^2 = \langle N^2 \rangle - \langle N^1 \rangle^2$; higher p give the hierarchical moments.

The pair count of Eq. (1) and the moment of Eq. (3) are related by integration over the cap geometry:

$$\langle N^1 \rangle^{\text{flavor}}(\theta; z_1, z_2) = \frac{1}{N_{P_1 \in z_1}} \int_0^\theta 2\pi\theta'\, d\theta'\, \frac{n^{\text{flavor}}_{\text{pairs}}(\theta'; z_1, z_2)}{d\theta'}. \tag{4}$$

This is the standard pair-count-to-counts-in-cells translation, applied per flavor.

2.4 kNN-CDF primitive: the Banerjee–Abel form

The third equivalent representation is the kNN cumulative distribution function per flavor:

$$P^{\text{flavor}}_{\ge k}(\theta; z_1, z_2) = \text{Prob}\big[N^{\text{flavor}}(\theta; z_2 \mid P_1 \text{ at } z_1) \ge k\big], \tag{5}$$

the probability that a P_1-center in bin z_1 has at least k P_2-neighbors in bin z_2 within angular cap θ.

By the Banerjee–Abel relation [4], the exact-count PDF is

$$P^{\text{flavor}}_{N=k}(\theta; z_1, z_2) = P^{\text{flavor}}_{\ge k}(\theta; z_1, z_2) - P^{\text{flavor}}_{\ge k+1}(\theta; z_1, z_2), \tag{6}$$

giving the probability of finding exactly k P_2-neighbors in the cap. The full CIC PDF is recovered for $k = 0, 1, 2, \ldots$ from the kNN ladder, with $P^{\text{flavor}}_{N=0} = 1 - P^{\text{flavor}}_{\ge 1}$.

2.5 Connection between the three representations

The pair-count, moment, and kNN-CDF representations contain the same information, organized differently. The bridges are:

Moments from kNN-CDFs. The moments of the CIC distribution follow from the kNN ladder via

$$\langle N^p \rangle^{\text{flavor}}(\theta; z_1, z_2) = \sum_{k=0}^{\infty} k^p\, P^{\text{flavor}}_{N=k}(\theta; z_1, z_2) = \sum_{k=0}^{\infty} k^p \big[P^{\text{flavor}}_{\ge k} - P^{\text{flavor}}_{\ge k+1}\big]. \tag{7}$$

For the mean specifically, the telescoping sum gives the simpler form

$$\langle N^1 \rangle^{\text{flavor}}(\theta; z_1, z_2) = \sum_{k=1}^{\infty} P^{\text{flavor}}_{\ge k}(\theta; z_1, z_2), \tag{8}$$

and likewise $\langle N^2 \rangle - \langle N^1 \rangle = \langle N(N-1) \rangle = 2\sum_{k \ge 2}(k-1) P^{\text{flavor}}_{\ge k}$.

Pair counts from moments. The differential pair count is the θ-derivative of the mean count:

$$\frac{d\, n^{\text{flavor}}_{\text{pairs}}(\theta; z_1, z_2)}{d\theta} = 2\pi\theta\, N_{P_1 \in z_1} \cdot \frac{\partial \langle N^1 \rangle^{\text{flavor}}(\theta; z_1, z_2)}{\partial \theta} / (2\pi\theta). \tag{9}$$

Equivalently, the cumulative pair count is $N_{P_1 \in z_1} \cdot \langle N^1 \rangle$.

kNN ladder from pair counts and joint structure. The kNN-CDF cannot in general be reconstructed from the mean count alone — it encodes additional information about the variance and higher moments of the cell-count distribution. The mapping from pair counts and the full joint distribution of cap counts to the kNN-CDF requires per-query bookkeeping at measurement time.

Choice of representation. For estimating the angular two- point function, pair counts are the natural primitive. For variance and counts-in-cells PDFs, moments are natural. For non-Gaussian probes and the full counts distribution, kNN-CDFs are natural. The same hierarchical pair-counting machinery produces all three: pair counts at the lowest level of the cascade traversal, moments by per-query accumulation of $\sum_q n_q^p$, and kNN-CDFs by per-query thresholding $\mathbb{1}[n_q \geq k]$.

2.6 What is encoded across all flavors

A complete data product is the set

$$\left\{ \langle N^p \rangle^{\text{flavor}}(\theta; z_1, z_2) \right\}_{\substack{p=1,2,\ldots,p_{\max} \\ \text{flavor} \in \{DD, DR, RD, RR\} \\ \theta \in \text{ angular bins} \\ z_1, z_2 \in \text{ redshift grid}}} \tag{10}$$

together with the equivalent kNN-CDF set $\{P^{\text{flavor}}_{\geq k}(\theta; z_1, z_2)\}_{k, \text{flavor}, \theta, z_1, z_2}$ (containing the same information in different organization). All standard observables are derived from this primitive via the bias-corrected estimators of Section 3.

3 Bias-corrected estimators

The four flavors carry related but distinct information. The natural, Landy–Szalay, and Hamilton combinations of them give bias-corrected estimators of the underlying density and clustering observables. We describe each in turn.

3.1 The natural estimator

The simplest combination uses only DD and RR. For the angular two- point function on the (z_1, z_2) shell pair, the Davis–Peebles natural estimator is

$$1 + \widehat{\xi}^{\text{nat}}(\theta; z_1, z_2) = \frac{\langle N^1 \rangle^{DD}(\theta; z_1, z_2)}{\langle N^1 \rangle^{RR}(\theta; z_1, z_2)}, \tag{11}$$

the ratio of the mean number of data neighbors around a data center to the mean number of references around a reference. It is unbiased to the extent that the reference catalog perfectly mirrors the angular and redshift selection of the data, but is sensitive to small departures from this assumption.

For one-point observables, the analogous natural estimator of mean density is $\langle N^1 \rangle^{RD} / \langle N^1 \rangle^{RR}$, comparing data density at reference centers to the geometric expectation. The Yuan, Zamora, Abel [10] 2D kNN framework implicitly uses these natural-estimator forms.

3.2 Landy–Szalay form

The Landy–Szalay combination of the four flavors removes selection- function bias to first order [1]:

$$\widehat{\xi}^{\mathrm{LS}}(\theta; z_1, z_2) = \frac{\langle N^1\rangle^{DD} - \langle N^1\rangle^{DR} - \langle N^1\rangle^{RD} + \langle N^1\rangle^{RR}}{\langle N^1\rangle^{RR}}. \tag{12}$$

The numerator is the connected pair density: uncorrelated contributions from selection variations cancel pairwise. The denominator normalizes to the geometric expectation.

For one-point statistics, the LS combinations of higher moments $\langle N^p\rangle^{\mathrm{flavor}}$ give bias-corrected estimates of the underlying counts-in-cells moments:

$$\widehat{\langle N^p\rangle}^{\mathrm{LS}}(\theta; z_1, z_2) = \frac{\langle N^p\rangle^{DD} - \langle N^p\rangle^{DR} - \langle N^p\rangle^{RD} + \langle N^p\rangle^{RR}}{\langle N^p\rangle^{RR}}. \tag{13}$$

From the bias-corrected moments, the variance follows as $\widehat{\sigma^2}^{\mathrm{LS}} = \widehat{\langle N^2\rangle}^{\mathrm{LS}} - (\widehat{\langle N^1\rangle}^{\mathrm{LS}})^2$, and the higher hierarchical moments S_p from the corresponding ratios of corrected cumulants. The dimensionless clustering variance with Poisson subtraction is

$$\widehat{\sigma^2_{\mathrm{clust}}}(\theta; z_1, z_2) = \frac{\widehat{\mathrm{Var}}}{\widehat{\langle N^1\rangle}^2} - \frac{1}{\widehat{\langle N^1\rangle}}. \tag{14}$$

3.3 Hamilton's form

Hamilton's estimator [2] for the two-point function is

$$\widehat{\xi}^{\mathrm{Ham}}(\theta; z_1, z_2) = \frac{\langle N^1\rangle^{DD} \cdot \langle N^1\rangle^{RR}}{\langle N^1\rangle^{DR} \cdot \langle N^1\rangle^{RD}} - 1. \tag{15}$$

Its multiplicative structure cancels overall normalization factors that affect the natural and LS estimators, and it is less sensitive to fluctuations in RR at large angular scales where RR counts are small. In the reference-redshift framework, RR is purely geometric — both endpoints sit at chosen reference grid points with exactly known positions — so RR has very small statistical noise. Hamilton's form then has noise dominated by the data-involving terms, which is the optimal allocation.

The analogous Hamilton combination for one-point moments is

$$\widehat{\langle N^p\rangle}^{\mathrm{Ham}}(\theta; z_1, z_2) = \frac{\langle N^p\rangle^{DD} \cdot \langle N^p\rangle^{RR}}{\langle N^p\rangle^{DR} \cdot \langle N^p\rangle^{RD}} - 1, \tag{16}$$

giving bias-corrected estimates of the underlying moments. For most cosmological applications, the Hamilton and LS forms agree at the level of statistical noise, and the choice between them is a matter of small-RR robustness vs. algebraic simplicity.

3.4 Bias correction of the kNN-CDF

The same LS or Hamilton combinations apply to the kNN-CDF directly:

$$\widehat{P_{\geq k}}^{\mathrm{LS}}(\theta; z_1, z_2) = \frac{P^{DD}_{\geq k} - P^{DR}_{\geq k} - P^{RD}_{\geq k} + P^{RR}_{\geq k}}{P^{RR}_{\geq k}}, \tag{17}$$

giving the bias-corrected kNN-CDF of the data, with the reference-grid contribution removed. The Banerjee–Abel relation (6) then gives the bias-corrected CIC PDF, and from that all bias-corrected moments. The kNN representation is therefore fully integrated into the four-flavor estimator framework.

3.5 Computational structure and the precomputed RR

The four flavors come from at most three pair-counting passes against the data tree:

1. *DD pass*: queries are data, search for data neighbors, with self-exclusion by index. Record $(\theta, z_{D_1}, z_{D_2})$ per pair.
2. *DR + RD pass*: queries are data, search for reference-grid neighbors. Each pair contributes once to DR (at $(z_D, z^{(r)})$) and once to RD (at $(z^{(r)}, z_D)$) by symmetry, so a single pass produces both. Reference points are stored in the same tree as data, distinguished by a flag.
3. *RR pass*: queries are references, search for references. Since both endpoints sit at chosen reference grid points with no intrinsic statistical content, this pass is purely geometric — it depends only on the angular footprint, the angular density of the reference catalog, and the chosen reference redshifts. **It can be precomputed once for any survey configuration and reused for all subsequent analyses.**

The precomputed RR is a substantial computational win. For a given survey footprint and reference grid, the RR moments and pair counts are fixed analytically and can be evaluated at high resolution without repeated tree traversal. Production analyses become single-tree-pass operations for DD and DR/RD, with RR looked up.

The reference catalog density is chosen for the desired statistical precision of the DR and RD terms, not constrained by the RR noise. For typical cosmological analyses, a reference density comparable to the data density per shell is sufficient: noise on bias-corrected estimators scales as $1/\sqrt{N_{\text{queries}}}$ independent of N_{data}.

4 The redshift derivative

Per-shell observables at a single redshift are theoretical objects on infinitesimal shells. The lightcone-native alternative is the redshift derivative, operationally accessible from finite differences across adjacent reference redshifts:

$$\widehat{\frac{\partial X}{\partial z}}(\theta; z + \Delta z/2) = \frac{\hat{X}(\theta; z + \Delta z) - \hat{X}(\theta; z)}{\Delta z}, \tag{18}$$

where $\hat{X}$ stands for any of the bias-corrected observables of Section 3 on the diagonal $z_1 = z_2 = z$, evaluated at reference grid points. The limit $\Delta z \to 0$ is the well-defined target; finite Δz is its discretization.

For a linear bias model with power-law evolution $b(z) = b_0(1+z)^p$ and matter clustering scaling with the linear growth $D(z)$,

$$\sigma^2(\theta; z) = b(z)^2\, D(z)^2\, \sigma_m^2\big(\theta\, D_M(z); z = 0\big), \tag{19}$$

and the logarithmic redshift derivative decomposes as

$$\frac{\partial \ln \sigma^2}{\partial \ln(1+z)} = \underbrace{2p}_{\text{bias}} + \underbrace{2(1+z)\frac{d\ln D}{dz}}_{\text{growth}} + \underbrace{n_{\text{eff}}\big(\theta D_M(z)\big) \cdot \frac{d\ln D_M}{d\ln(1+z)}}_{\text{geometry}}, \tag{20}$$

where $n_{\text{eff}}(r) = d\ln\sigma_m^2/d\ln r$ is the effective slope of the matter variance. Each term is well-defined and individually constrainable:

- The bias-evolution term $2p$ is constant in $\log(1+z)$ at each θ, isolating the bias evolution index as a directly measurable quantity.
- The growth term is smooth and monotonic, asymptoting in matter domination and carrying $\Omega_\Lambda(z)$ information.
- The geometric term carries the BAO standard ruler: the function $n_{\rm eff}(r)$ has a localized feature near $r \sim r_{\rm BAO}$, and as z varies, this feature appears in $\partial \ln \sigma^2 / \partial \ln(1+z)$ at the angular scale satisfying $\theta D_M(z) = r_{\rm BAO}$, tracing a curve in $(\log\theta, \log(1+z))$ space.

5 Information equivalence with $\xi(r)$ and multipole recovery

5.1 Fisher information

The lightcone-native primitive is a linear functional of the underlying clustering field. The Fisher information on a parameter Θ from any complete linear basis of $\xi(r;z)$ is the same, under Gaussian covariance. The matched-filter bound,

$$F^{\rm max}_{\Theta\Theta} = \frac{V_{\rm shell}}{2\pi^2} \int dk\, k^2 \, \frac{(\partial P/\partial\Theta)^2}{[P(k) + 1/\bar{n}]^2}, \tag{21}$$

is achieved by any estimator that linearly combines $P(k)$ measurements optimally. Both $\xi(r;z)$ at multiple separations and per-shell $\sigma^2(\theta;z)$ at multiple angular scales are such combinations, related by linear integral transforms.

We verified the equivalence numerically using the `syren-new` symbolic-regression emulator [8] for the linear matter power spectrum and an Eisenstein–Hu no-wiggle template [7] for the BAO signal model. For survey parameters representative of an LRG sample ($V = 10^9\,(h^{-1}{\rm Mpc})^3$, $\bar{n} = 3\times10^{-4}\,(h\,{\rm Mpc}^{-1})^3$, $z_{\rm eff} = 0.5$), the Fisher information on the BAO dilation parameter $\alpha_{\rm BAO}$:

Estimator	$F(\alpha_{\rm BAO})$	Efficiency
Matched filter (Cramér–Rao bound)	5350	100%
Binned $\xi(r)$, full Gaussian covariance, 130 bins	≈ 5350	$\approx 100\%$
Top-hat shell bank, 50 shells of $\Delta r = 10\,h^{-1}$Mpc	5270	98.5%

A dense bank of per-shell variance measurements recovers all the information available in $\xi(r)$. The two representations are information-equivalent.

5.2 Recovering traditional multipoles from gridded values

The conventional analysis decomposes the redshift-space correlation function $\xi(s,\mu)$ into Legendre multipoles

$$\xi_\ell(s) = \frac{2\ell+1}{2} \int_{-1}^{1} d\mu\, L_\ell(\mu)\, \xi(s,\mu), \tag{22}$$

where s is the redshift-space separation and μ is the cosine of the angle between the pair-separation vector and the line of sight. These multipoles — typically ξ_0 (monopole), ξ_2 (quadrupole), ξ_4 (hexadecapole) — are the standard data product for cosmological inference.

The lightcone-native primitive $\tilde{\xi}(\theta; z_1, z_2)$, evaluated on the diagonal subset $z_1 = z_2 = z$ and on a (z_1, z_2) grid with small $|z_1 - z_2|$, encodes the same information as $\xi(s,\mu)$ at the effective redshift, related by an explicit cosmology-dependent mapping.

To derive multipoles from the gridded values: given a fiducial cosmology, the mapping from observable separations (θ_{12}, z_1, z_2) to redshift-space coordinates (s, μ) is

$$r_\perp = D_M(z_{\rm mid})\,\theta_{12}, \qquad r_\parallel = \frac{c\,(z_2 - z_1)}{H(z_{\rm mid})}, \tag{23}$$

with $z_{\rm mid} = (z_1 + z_2)/2$. From these,

$$s = \sqrt{r_\perp^2 + r_\parallel^2}, \qquad \mu = \frac{r_\parallel}{s}. \tag{24}$$

The traditional multipoles then follow from the Legendre projection (22), with the gridded primitive $\hat{\tilde{\xi}}^{\rm LS}(\theta_{12}; z_1, z_2)$ supplying the underlying $\xi(s, \mu)$ via the substitution (24).

This makes the framework's role in conventional analyses explicit:

- The measurement of $\tilde{\xi}(\theta; z_1, z_2)$ on a grid is the fundamental observation, made without invoking any fiducial cosmology.
- Cosmology enters at the comparison step via Eq. (23), which defines the (s, μ) coordinates of each (θ, z_1, z_2) gridpoint.
- Multipoles $\xi_\ell(s)$ are then computed by projecting the gridded primitive against the chosen (s, μ) mapping. Different cosmologies map the same data vector to different multipoles; the likelihood compares predicted to measured multipoles in coefficient space.

This is the cleanest realization of the Alcock–Paczynski test: $\hat{\tilde{\xi}}(\theta; z_1, z_2)$ is cosmology-independent; the AP distortion appears as a coordinate transformation when one passes to (s, μ). The lightcone-native pipeline supplies all the information needed to compute the conventional multipoles for any chosen cosmology; no recomputation of the data vector is required when the cosmology parameter changes.

6 Higher-order statistics and halo mass thresholds

The same four-flavor framework extends to higher-order moments and mass-thresholded subsets. The third and fourth moments $\langle N^3\rangle^{\rm flavor}$ and $\langle N^4\rangle^{\rm flavor}$, computed by the same per-query accumulator that gives the variance, provide bias-corrected estimates of S_3 (skewness) and S_4 (kurtosis) of the cone-shell counts. The kNN- CDF ladder $P^{\rm flavor}_{\geq k}$ contains the same information in a different organization, with the exact-count PDF $P_{N=k}$ available from adjacent differences.

For halo catalogs from N-body simulations, the framework gains a mass- threshold axis $M_{\rm min}$. The primitive extends to

$$\langle N^p\rangle^{\rm flavor}(\theta; z_1, M_{\rm min,1}, z_2, M_{\rm min,2}), \tag{25}$$

with both query and neighbor populations restricted to halos with $M_h > M_{\rm min}$. A scan over $M_{\rm min}$ values gives all threshold measurements in a single tree-traversal pass: as each neighbor is visited, it contributes to the accumulator at all thresholds $M_{\rm min} \leq M_h$.

The diagonal $M_{\rm min,1} = M_{\rm min,2}$ is the autocorrelation of a single mass-selected sample. The off-diagonal $M_{\rm min,1} \neq M_{\rm min,2}$ is the cross-correlation between mass-selected populations:

$$\bar{n}^{DD}(\theta; z, M_{\rm min,1}, M_{\rm min,2}) \propto 1 + b(M_{\rm min,1})\, b(M_{\rm min,2})\, \xi_m(\theta D_M(z); z), \tag{26}$$

giving the bias matrix as a directly observable quantity. The strong differential response of $\bar{n}$ and σ^2 to $M_{\rm min}$ provides tight cosmological leverage through halo abundance and bias, breaking degeneracies present in any single-threshold measurement.

7 Photometric redshifts and z-resolution

When per-galaxy redshift uncertainties σ_z approach the reference grid spacing, the natural treatment is soft assignment: each galaxy contributes to multiple reference-grid bins with weights

$$p_i^{(s)} = \int_{z_s^{\rm lo}}^{z_s^{\rm hi}} \mathcal{N}\big(z; \tilde{z}_i, \sigma_z^{(i)}\big)\, dz, \tag{27}$$

where $\tilde{z}_i$ is the observed redshift and the integral runs over the bin. The accumulator generalizes to weighted pair counts and weighted moments; the framework operates at the resolution set by the survey's redshift quality automatically.

The reference grid spacing is naturally set to $\Delta z \approx \sigma_z$, beyond which finer binning carries no information. Reference points have zero redshift uncertainty by construction — they are theoretical anchors — so the RR flavor is unaffected by photo-z errors. The DD flavor inherits the full per-galaxy uncertainty; the DR and RD flavors have uncertainty only on one side. This natural asymmetry can be exploited in survey design and analysis.

8 Cosmology enters only as a forward model

In this framework, cosmology enters at exactly two well-defined places:

1. *Mapping observables to physical scales for interpretation.* The angular scale θ at redshift z corresponds to a spatial scale $r = D_M(z)\theta$ only after specifying $D_M(z)$. Different cosmologies map the same primitive $\tilde{\xi}(\theta; z_1, z_2)$ to different physical scales. The data vector is unchanged.

2. *Forward-modeling the predicted observables.* Given a cosmology, growth, bias model, RSD model, and (for galaxy surveys) HOD, one computes the predicted four-flavor moments and compares to data on the $(\log\theta, \log(1+z_1), \log(1+z_2))$ grid. Traditional multipoles $\xi_\ell(s)$ are computed by projection at this stage, not before.

The Alcock–Paczynski test in this picture is not a distortion to be fit out: it is the statement that the mapping $\theta \to r = D_M(z)\theta$ depends on cosmology, and the data constrain $D_M(z)$ through the redshift dependence of the clustering pattern. RSD live in the off- diagonal $z_1 \neq z_2$ structure of $\tilde{\xi}(\theta; z_1, z_2)$, computed from the same forward model.

9 Discussion and outlook

The proposal is not novel in any individual ingredient. Tomographic angular clustering is standard for photometric surveys [6]. The Alcock–Paczynski caution is well known [5]. The kNN-CDF formalism and the relation to cone-shell counts-in-cells have been developed extensively [4, 9, 10, 11]. What is new is a *unified* treatment in which all clustering statistics — mean count, two-point function, counts-in-cells PDF, higher-order moments, RSD, mass-threshold cross-correlations — emerge as Landy–Szalay or Hamilton combinations of four-flavor primitive moments on $(\log\theta, \log(1+z_1), \log(1+z_2))$, computed in a single pass with cosmology-free data vectors and a precomputable RR contribution.

The framework is well-suited to spectroscopic surveys with broad redshift coverage (DESI, Euclid, Roman), to photometric surveys (LSST), and to cross-survey analyses where the same machinery operates on different $\bar{n}(\Omega, z)$ but the same coordinate primitive. The `morton_cascade`

pair-counting infrastructure, with its $(\Omega, \log(1+z))$ Morton ordering, provides hierarchical traversal natively; the natural z-locality enables per-shell-pair pruning of tree subtrees for fast per-shell statistics.

Several extensions are natural and reserved for follow-up. A basis- function decomposition gives an alternative finite-dimensional representation in which cosmological parameters enter through closed-form expressions for the basis coefficients. Density-conditioned and quantile- based variants extract additional non-Gaussian information with shot- noise efficiency. A demonstration on lightcone simulations [9], comparing the lightcone-native pipeline against conventional binned estimators on the same Abacus-Summit catalogs with realistic selection effects, would validate the framework on survey- quality data.

A Practical recommendations for production binning

This appendix records concrete binning choices for production deployment of the framework on spectroscopic surveys (DESI, Euclid, Roman). The recommendations balance information preservation with shot-noise per bin and downstream rebinning flexibility. The data product is intended to serve as a survey-archive-quality clustering primitive that can be coarsened or projected to any analysis question without recomputation.

A.1 Two-point function: pair counts on a fine (θ, z_1, z_2) grid

Pair counts are stored on a uniform grid:

- *Angular:* 100 logarithmic bins in $\theta \in [0.05, 20]$, giving $\Delta \log_{10} \theta \approx 0.026$. The lower edge is set by the shot-noise threshold (sub-arcminute scales contribute $\langle N \rangle < 1$ per bin even for dense surveys); the upper edge covers all BAO and broad-scale clustering features.
- *Redshift axes:* 100 bins each in $\log(1 + z_1)$ and $\log(1 + z_2)$, spanning the 5%–95% quantile range of the survey's $N(z)$ distribution. The trimming of tails avoids edge effects from the selection function. Uniform spacing in $\log(1 + z)$ gives natural lightcone binning (Section 4).
- *Flavors and asymmetry:* All four flavors $\{DD, DR, RD, RR\}$ are computed. DD and RR are symmetric under $z_1 \leftrightarrow z_2$ and stored on the lower triangle $z_1 \leq z_2$ only. DR and RD are not symmetric and are stored on the full z_1, z_2 plane (or equivalently, only DR is stored with RD recovered by reflection).

The pair-count primitive total size for a single moment order is $100 \times 100 \times 100 \times 8$ bytes = 8 MB per flavor on the full plane, $\sim$ 4 MB per symmetric flavor. The full $\{DD, DR, RD, RR\}$ set is $\sim$ 30 MB.

A.2 Counts-in-cells and kNN statistics: separate grids for R and D centered measurements

The counts-in-cells and kNN statistics use specialized reference grids chosen to balance signal-to-noise per query population.

R-centered kNN/CIC. 9 reference redshifts placed at the deciles of the data $N(z)$ distribution (cumulative quantiles 0.1, 0.2, ..., 0.9). Random query points are placed at exactly each reference redshift, sampled uniformly on the sky within the survey footprint (HEALPix centers or Poisson sampling at angular density comparable to $\bar{n}$).

The decile spacing places references where the galaxy density is highest in $N(z)$-space, giving each measurement comparable shot-noise structure. The R-centered measurements probe the field as seen from generic random angular positions: voids, intermediate-density regions, and cluster outskirts contribute equally by angular area. This is the natural configuration for measuring the angular-position-averaged variance, hierarchical moments, and the selection-function-like baseline.

D-centered kNN/CIC. 91 reference redshift edges placed at the quantiles $\{0.095, 0.105, 0.115, \ldots, 0.895, 0.905\}$ of the data $N(z)$, defining 90 reference-redshift bins each containing 1% of the galaxies. Query points are the data galaxies themselves, restricted to those whose redshift quantile falls within the bulk range $[0.095, 0.905]$. The 90 bins each have approximately equal query counts.

The data-centered measurements probe the field as seen from *galaxy positions*: the queries sit inside the clustered field where the small-scale signal is most concentrated. The fine 1%-quantile spacing preserves the redshift resolution of the data themselves. Trimming the outer tails (the 5% on each end of $N(z)$) suppresses edge effects from the selection function and from declining number density at the survey boundaries.

kNN ladder depth. $k_{\max} = 302$ for both R- and D-centered measurements. This is generous: at the diagonal $z_n = z_q$ on the deeper R-shells, $\langle N \rangle$ exceeds 100 at the largest angular scales, and $P_{\geq k}$ saturates near $k \sim 3\langle N \rangle$. The chosen $k_{\max}$ captures the full distribution out to the $\sim$ 99th-percentile for all expected query densities.

A.3 Why R and D centered measurements have different grids

The two flavors carry complementary information and have different sampling properties. R-centered queries are uniform on the sky — their density is set by the user. D-centered queries are the data themselves — their density is set by the survey. The natural reference-redshift grids therefore differ:

- R-centered uses a *coarse* grid (9 redshifts) because each reference defines an independent measurement and there is little gain from oversampling: the R-centered field is dominated by large-scale structure, varying smoothly with redshift.
- D-centered uses a *fine* grid (90 redshift bins) because each data galaxy carries its own redshift and we want to preserve the intrinsic redshift resolution. Coarsening would discard information about how clustering correlates with the precise redshift of the central galaxy.

This asymmetry mirrors the natural noise structure: R-centered noise scales as $1/\sqrt{N_R}$ at each reference, while D-centered noise scales as $1/\sqrt{N_D \cdot (\text{fraction in bin})}$, which is minimized by splitting the data into many fine-quantile bins.

A.4 Storage and downstream rebinning

The total primary data product for a typical spectroscopic survey is approximately:

- Pair counts (all four flavors, four moments $p = 1, \ldots, 4$): $\sim$ 130 MB

- R-centered CIC and kNN (9 references $\times$ 100 θ bins $\times$ 9 neighbor shells $\times$ 302 k-values, four flavors): $\sim$ 80 MB
- D-centered CIC and kNN (90 references $\times$ 100 θ bins $\times$ 90 neighbor shells $\times$ 302 k-values, four flavors): $\sim$ 2.5 GB
- Precomputed RR tables and selection-function maps: $\sim$ 100 MB

Total: approximately 3 GB, dominated by the D-centered kNN ladder. HDF5 storage with chunking along the redshift axes allows downstream analyses to load slices without reading the whole archive.

Downstream rebinning is straightforward:

- *Coarsen redshift*: sum adjacent (z_1, z_2) cells in the pair-count primitive, weighted by query density per cell.
- *Project to traditional multipoles*: apply the cosmology- dependent $(\theta, z_1, z_2) \to (s, \mu)$ mapping (Section 5) and Legendre-project.
- *Restrict to BAO range*: slice on $\theta D_M(z) \in [80, 130]\, h^{-1}$Mpc and study cross-shell behavior.
- *Cosmology variation*: recompute the $(\theta, z_1, z_2) \to (s, \mu)$ mapping per cosmology; reproject. The primary data product is unchanged; only the projection varies.

A.5 Compute cost estimate

For a DESI-scale catalog ($N_d \sim 10^7$, $N_R \sim N_d$ per reference redshift):

- DD pass: one tree walk with self-exclusion, $\sim$ 10 minutes on a 64-core node using a Morton-cascade-based implementation.
- DR+RD pass (joint): one tree walk, $\sim$ 10 minutes.
- RR pass: purely geometric, precomputable, $\sim$ 1 minute.

Total: approximately 20–30 minutes of compute per analysis on a single modern node. The result is a complete clustering archive suitable for all downstream cosmological inference on the survey.

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